Hierarchical Active Inference for Trajectory Tracking

A Two-Tier Model: Continuous Low-Level Agent + Discrete High-Level Agent
with Efference Copy, Precision Modulation, and Cross-Modal Surprise

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1 Motivation and Overview

The single-level continuous agent (documented separately) successfully tracks a 3-D trajectory under gravity but has no mechanism to respond to unexpected events. The hierarchical extension adds a **discrete high-level agent** that monitors for surprising cross-modal stimuli and — through *precision modulation* — transiently perturbs the low-level controller.

The model is designed to reproduce a specific experimental signature:

Target phenomenology (from experiments).

- Sudden, task-irrelevant stimuli produce **triphasic kinematic perturbations** (decrease–increase–decrease in acceleration) aligned with the dominant force axis.
- Perturbations are **larger during movement** than during stationary holding.
- Electrocortical responses (vertex potentials) are **smaller during movement** — an anti-correlation.

One mechanism must account for both signatures simultaneously.

2 Architecture

The model has two tiers that exchange information every timestep.

2.1 State and action spaces (low level)

Unchanged from the single-agent model:

$$\boldsymbol{x} = [p_1, p_2, p_3, v_1, v_2, v_3]^\top \in \mathbb{R}^6, \qquad \boldsymbol{a} = [u_1, u_2]^\top \in \mathbb{R}^2, \quad u_3 \equiv 0. \tag{1}$$

2.2 State space (high level)

The discrete agent maintains a categorical belief over three states:

$$s \in \{\underbrace{\text{TRACK}}_{0}, \underbrace{\text{SURPRISED}}_{1}, \underbrace{\text{RECOVER}}_{2}\}, \qquad \boldsymbol{s}_t \in \Delta^2 \ \text{(probability simplex)}. \tag{2}$$

2.3 Coupling signals

Signal	Direction	Definition
Efference copy	low → high	$e(t) = \lVert \boldsymbol{a}(t) \rVert, \quad \bar{e}(t) = e(t)/a_{\max} \in [0,1]$
Precision scale	high → low	$\boldsymbol{\Pi}_s^{\text{low}}(t) = \sigma_{\text{low}}(\gamma(t))\, \boldsymbol{\Pi}_s^{\text{base}}$

3 High-Level Discrete Agent

3.1 Observation model

The high-level agent receives a binary cross-modal observation $o_t \in \{0 \text{ (no stim)}, 1 \text{ (stim)}\}$. The likelihood matrix $\mathbf{A} \in [0,1]^{3\times 2}$ encodes that Surprised is the only state consistent with stimulus presence:

$$\mathbf{A} = \begin{pmatrix} 0.95 & 0.05 \\ 0.05 & 0.95 \\ 0.90 & 0.10 \end{pmatrix}, \quad A_{s,o} = p(o_t = o \mid s_t = s). \tag{3}$$

3.2 Transition model — efference modulation

The transition matrix $\mathbf{B}(\bar{e})$ is modulated by the normalised efference copy $\bar{e}(t) \in [0, 1]$:

$$p(\text{TRACK} \to \text{SURPRISED}) = p_0 + \alpha_B\, \bar{e}, \qquad p(\text{SURP} \to \text{RECOVER}) = q_0 + \alpha_B\, \bar{e}, \qquad p(\text{REC} \to \text{TRACK}) = r \tag{4}$$

> **Interpretation.** High motor activity (large $\bar{e}$) makes the system easier to destabilise (Track→Surprised more likely) and accelerates recovery (Surprised→Recover→Track faster). This is the mechanism by which the *same* stimulus produces a larger kinematic effect during movement.

3.3 Belief update — softmax over log-likelihoods

Following standard discrete AIF, the belief is updated in two steps:

Step 1 — Prior prediction.

$$\boldsymbol{s}_{\text{prior}} = \mathbf{B}(\bar{e})\, \boldsymbol{s}_{t-1}, \qquad \boldsymbol{s}_{\text{prior}} \in \Delta^2. \tag{5}$$

Step 2 — Bayesian update (posterior).

$$\tilde{s}_t(s) \propto A_{s,\,o_t} \cdot s_{\text{prior}}(s), \qquad \boldsymbol{s}_t = \frac{\tilde{\boldsymbol{s}}_t}{\|\tilde{\boldsymbol{s}}_t\|_1}. \tag{6}$$

This is equivalent to softmax over log-likelihoods:

$$\ln \boldsymbol{s}_t = \ln \boldsymbol{s}_{\text{prior}} + \ln \mathbf{A}_{:,\,o_t} + \text{const}. \tag{7}$$

3.4 Surprise and high-level free energy

$$p(o_t) = \sum_s A_{s,o_t}\, s_{\text{prior}}(s) \quad \text{(marginal likelihood)}, \tag{8}$$

$$\xi_t = -\ln p(o_t) \quad \text{(surprise)}, \tag{9}$$

$$\mathcal{F}_{\text{high},t} = \underbrace{D_{\text{KL}}[\boldsymbol{s}_t \| \boldsymbol{s}_{\text{prior}}]}_{\text{belief update cost}} + \underbrace{\xi_t}_{\text{surprise}}. \tag{10}$$

Both ξ_t and $\mathcal{F}_{\text{high},t}$ are stored as **candidate proxies for the vertex potential** (see Sec. 6).

4 Precision Coupling

4.1 Surprise pulse $\gamma(t)$

The high-level surprise ξ_t drives a leaky-integrator pulse:

$$\dot{\gamma} = -\frac{\gamma}{\tau_\gamma} + \lambda_\gamma\, \xi_t, \qquad \gamma(0) = 0, \quad \gamma \geq 0. \tag{11}$$

The pulse rises sharply at stimulus onset and decays exponentially with time constant τ_γ.

4.2 Low-level precision modulation

$$\boxed{\mathbf{\Pi}_s^{\text{low}}(t) = \frac{1}{1+\gamma(t)}\,\mathbf{\Pi}_s^{\text{base}},} \tag{12}$$

where $\mathbf{\Pi}_s^{\text{base}}$ is the baseline 6×6 precision matrix (z-entries zero, as before). When $\gamma \uparrow$, low-level sensory precision drops: the agent *loosens its grip* on the trajectory, generating the kinematic perturbation.

4.3 High-level precision modulation

$$\boxed{\Pi_s^{\text{high}}(t) = \frac{\Pi_s^{\text{high,base}}\,(1+\gamma(t))}{1+\alpha_{\text{eff}}\,\bar{e}(t)},} \tag{13}$$

where $\alpha_{\text{eff}} > 0$ controls efference suppression. When $\bar{e}(t)$ is large (active movement), Π_s^{high} is suppressed at baseline — representing predictive motor suppression.

Independence of the two precisions. $\mathbf{\Pi}_s^{\text{low}}$ and Π_s^{high} are *not* coupled by a conservation law. They happen to move in opposite directions at stimulus onset because both are functions of $\gamma(t)$, but through independent pathways.

5 Low-Level Continuous Agent

The low-level equations are identical to the single-agent version except that $\mathbf{\Pi}_s^{\text{low}}$ is now time-varying.

5.1 Belief update

$$\dot{\boldsymbol{\mu}} = \boldsymbol{f}_{\text{model}}(\boldsymbol{\mu}, \boldsymbol{a}) + \kappa_\mu\big(\mathbf{\Pi}_s^{\text{low}}(t)\,\boldsymbol{\varepsilon}_s - \mathbf{\Pi}_p\,\boldsymbol{\varepsilon}_p\big). \tag{14}$$

5.2 Precision-modulated action

The PD controller is scaled by the low-level precision:

$$u_1 = \sigma(\gamma)\,m_1\big[K_P(p_1^* - \mu_{p_1}) + K_D(v_1^* - \mu_{v_1})\big], \tag{15}$$

$$u_2 = \sigma(\gamma)\,m_2\big[K_P(p_2^* - \mu_{p_2}) + K_D(v_2^* - \mu_{v_2})\big] + m_2 g, \tag{16}$$

where $\sigma(\gamma) = 1/(1+\gamma) \in (0, 1]$ is the precision scale.

Origin of the triphasic perturbation. At stimulus onset γ rises sharply, so $\sigma(\gamma) \to 0$:

1. **Phase 1 (decrease):** PD forces drop — the agent loses grip.
2. **Phase 2 (increase):** γ begins to decay; position error has grown, so forces overshoot as the controller re-engages.
3. **Phase 3 (decrease):** Normal convergence to $\boldsymbol{x}^*$.

The perturbation is automatically *force-aligned* because it acts through the force channel, not through a spatial direction vector.

5.3 Low-level free energy

$$\mathcal{F}_{\text{low}} = \tfrac{1}{2}\boldsymbol{\varepsilon}_p^\top \mathbf{\Pi}_p \boldsymbol{\varepsilon}_p + \tfrac{1}{2}\boldsymbol{\varepsilon}_s^\top \mathbf{\Pi}_s^{\text{low}}(t)\boldsymbol{\varepsilon}_s. \tag{17}$$

6 Vertex Potential Proxies

The vertex potential (N100/P300 complex in EEG) reflects neural surprise. Two candidate proxies are generated by the model:

Proxy	**Expression**	**Interpretation**
$\mathrm{VP}_{\Pi\varepsilon}$	$\Pi_s^{\mathrm{high}}(t)\ \left\lVert \boldsymbol{\varepsilon}_s^{\mathrm{high}}(t) \right\rVert$	Precision-weighted prediction error
$\mathrm{VP}_{\mathcal{F}}$	$\mathcal{F}_{\mathrm{high},t}$ (Eq. 10)	Full high-level free energy

Both are stored in the simulation output so they can be regressed against empirical EEG amplitudes to determine which is the better neural correlate.

Predicted anti-correlation. During movement $\bar{e}(t)$ is large, suppressing Π_s^{high} via the denominator of Eq. (13), so both proxies are attenuated at stimulus onset. The kinematic disruption is large because $\boldsymbol{a}$ was large before the reset. During stationary $\bar{e} \approx 0$, so Π_s^{high} is at its maximum and both proxies are large — but the kinematic disruption is small because the forces being reset were near zero.

7 Algorithm

Algorithm 1 Hierarchical AIF — Two-Tier Simulation Loop

Require: All parameters; likelihood matrix $\mathbf{A}$; noise arrays

Init: $\boldsymbol{x}_{\text{real}}, \boldsymbol{\mu} \leftarrow \boldsymbol{x}_0$; $\boldsymbol{s} \leftarrow [0.95, 0.025, 0.025]$; $\gamma \leftarrow 0$; $\boldsymbol{a}_{\text{prev}} \leftarrow \mathbf{0}$

for $k = 0, \ldots, N-1$ **do**

  $t \leftarrow k\,\Delta t$; $\boldsymbol{x}^*(t) \leftarrow \textsc{DesiredState}(t)$

  — **Efference copy** —

  $e \leftarrow \|\boldsymbol{a}_{\text{prev}}\|$; $\bar{e} \leftarrow e/a_{\max}$

  — **High-level: discrete belief update** —

  $o_t \leftarrow \mathbb{1}[\text{stimulus active at } t]$

  $\mathbf{B} \leftarrow \textsc{BuildTransitionMatrix}(\bar{e})$

  $\boldsymbol{s}_{\text{prior}} \leftarrow \mathbf{B}\,\boldsymbol{s}$

  $p(o_t) \leftarrow \mathbf{A}_{:,\,o_t}^{\top} \boldsymbol{s}_{\text{prior}}$; $\xi \leftarrow -\ln p(o_t)$

  $\boldsymbol{s} \leftarrow \mathbf{A}_{:,\,o_t} \odot \boldsymbol{s}_{\text{prior}}$; normalise $\boldsymbol{s}$

  $\mathcal{F}_{\text{high}} \leftarrow D_{\text{KL}}[\boldsymbol{s} \| \boldsymbol{s}_{\text{prior}}] + \xi$

  — **Precision pulse update** —

  $\gamma \leftarrow \gamma + \left(-\gamma/\tau_\gamma + \lambda_\gamma\,\xi\right)\Delta t$; $\gamma \leftarrow \max(\gamma, 0)$

  $\sigma \leftarrow 1/(1+\gamma)$ ▷ low-level scale

  $\Pi_s^{\text{high}} \leftarrow \Pi_s^{\text{high,base}}\,(1+\gamma)\,/\,(1+\alpha_{\text{eff}}\,\bar{e})$ ▷ high-level precision

  — **Low-level: perception** —

  Build $\boldsymbol{\Pi}_s^{\text{low}} \leftarrow \sigma\,\boldsymbol{\Pi}_s^{\text{base}}$

  Observe $\boldsymbol{y}_{\text{obs}}$ (x-y only, z-entries $\leftarrow \boldsymbol{\mu}_z$)

  $\boldsymbol{\varepsilon}_p \leftarrow \boldsymbol{\mu} - \boldsymbol{x}^*$; $\boldsymbol{\varepsilon}_s \leftarrow \boldsymbol{y}_{\text{obs}} - \boldsymbol{\mu}$

  $\mathcal{F}_{\text{low}} \leftarrow \frac{1}{2}(\boldsymbol{\varepsilon}_p^{\top}\boldsymbol{\Pi}_p\boldsymbol{\varepsilon}_p + \boldsymbol{\varepsilon}_s^{\top}\boldsymbol{\Pi}_s^{\text{low}}\boldsymbol{\varepsilon}_s)$

  — **Low-level: action** —

  $u_1 \leftarrow \sigma\,m_1[K_P(\mu_{p_1}^* - \mu_{p_1}) + K_D(\mu_{v_1}^* - \mu_{v_1})]$

  $u_2 \leftarrow \sigma\,m_2[K_P(\mu_{p_2}^* - \mu_{p_2}) + K_D(\mu_{v_2}^* - \mu_{v_2})] + m_2 g$

  Clip $u_1, u_2 \in [-a_{\max}, a_{\max}]$; $u_3 \leftarrow 0$; $\boldsymbol{a}_{\text{prev}} \leftarrow \boldsymbol{a}$

  — **World update** —

  $\boldsymbol{x}_{\text{real}} \leftarrow \boldsymbol{x}_{\text{real}} + \boldsymbol{f}_{\text{proc}}(\boldsymbol{x}_{\text{real}}, \boldsymbol{a}, \boldsymbol{\omega}_k)\,\Delta t$

  — **Belief update** —

  $\boldsymbol{\mu} \leftarrow \boldsymbol{\mu} + \left[\boldsymbol{f}_{\text{model}}(\boldsymbol{\mu}, \boldsymbol{a}) + \kappa_\mu(\boldsymbol{\Pi}_s^{\text{low}}\boldsymbol{\varepsilon}_s - \boldsymbol{\Pi}_p\boldsymbol{\varepsilon}_p)\right]\Delta t$

  — **Vertex potential proxies** —

  $\text{VP}_{\Pi\varepsilon} \leftarrow \Pi_s^{\text{high}}\,\|\boldsymbol{\varepsilon}_{s,xy}\|$; $\text{VP}_{\mathcal{F}} \leftarrow \mathcal{F}_{\text{high}}$

  Store all quantities

end for

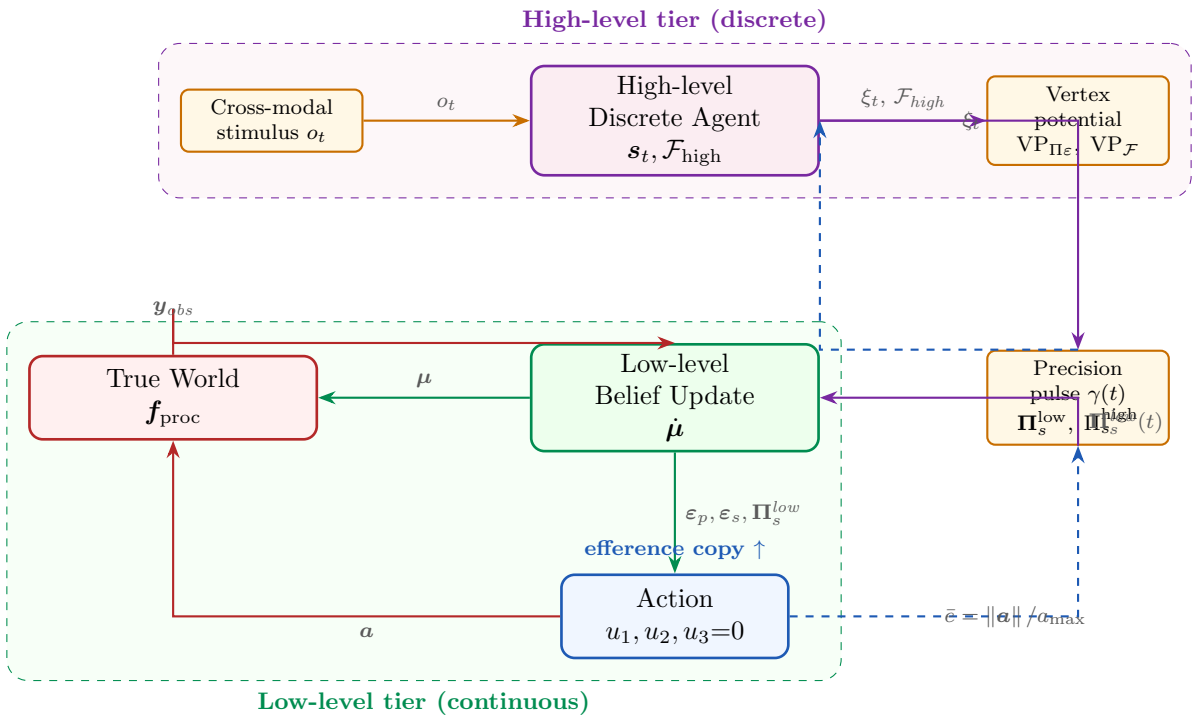

Figure 1: Two-tier hierarchical AIF architecture. Purple: high-level discrete agent. Green: low-level continuous agent and true world. Blue dashed: efference copy pathway. Orange: cross-modal stimulus input and precision pulse.

8 System Diagram

9 Predicted Anti-Correlation — Formal Derivation

Let t^* denote the stimulus onset. Define:

$$\Delta_{\text{kin}}(t^*) = \max_{t>t^*} \|\boldsymbol{a}(t) - \boldsymbol{a}(t^*-)\| \quad \text{(peak kinematic perturbation)}, \tag{18}$$

$$\text{VP}(t^*) = \text{either proxy evaluated at } t = t^*. \tag{19}$$

Movement condition. $\bar{e}(t^*) = \bar{e}_M > 0$, $\|\boldsymbol{a}(t^*)\| = a_M > 0$.

$$\text{VP}_{\Pi\varepsilon}^M = \frac{\Pi_s^{\text{high,base}}\,(1+\gamma^+)}{1+\alpha_{\text{eff}}\,\bar{e}_M} \cdot \left\|\boldsymbol{\varepsilon}_{s,xy}^+\right\|, \tag{20}$$

$$\Delta_{\text{kin}}^M \approx a_M \cdot (1-\sigma(\gamma^+)) = a_M\,\frac{\gamma^+}{1+\gamma^+}. \tag{21}$$

Stationary condition. $\bar{e}(t^*) \approx 0$, $\|\boldsymbol{a}(t^*)\| \approx 0$.

$$\text{VP}_{\Pi\varepsilon}^S = \Pi_s^{\text{high,base}}\,(1+\gamma^+) \cdot \left\|\boldsymbol{\varepsilon}_{s,xy}^+\right\|, \tag{22}$$

$$\Delta_{\text{kin}}^S \approx 0 \cdot (\cdots) \approx 0. \tag{23}$$

Comparing Eqs. (20) and (22):

$$\frac{\text{VP}^M}{\text{VP}^S} = \frac{1}{1+\alpha_{\text{eff}}\,\bar{e}_M} < 1, \tag{24}$$

and from Eqs. (21) and (23):

$$\Delta_{\text{kin}}^M > \Delta_{\text{kin}}^S. \tag{25}$$

Anti-correlation from a single mechanism. The efference suppression term $1/(1+\alpha_{\text{eff}}\bar{e})$ in Π_s^{high} simultaneously:

- decreases the vertex potential proxy during movement ($\text{VP}^M < \text{VP}^S$),
- increases transition probability to SURPRISED, amplifying γ, which amplifies Δ_{kin}.

No conservation law is required. The two effects emerge from the same scalar $\bar{e}(t)$ entering two independent equations.

10 Parameter Summary

Table 1: Hierarchical extension parameters (low-level unchanged from Table 1).

Symbol	Description	Default	Role
α_B	Efference→transition modulation	2.0	High-level B matrix
α_{eff}	Efference→ Π_s^{high} suppression	0.05	Vertex potential attenuation
τ_γ	Surprise pulse decay time	0.3 s	Perturbation duration
λ_γ	Surprise→ γ gain	1.5	Perturbation amplitude
$\Pi_s^{\text{high,base}}$	Baseline high-level precision	10.0	VP baseline
p_0	Base TRACK→SURP probability	0.05	Spontaneous surprisability
q_0	Base SURP→REC probability	0.40	Recovery speed baseline
r_0	Base REC→TRACK probability	0.30	Re-engagement speed baseline

11 Open Questions for Model Testing

Q1. Which vertex potential proxy fits EEG data better? $\text{VP}_{\Pi_\varepsilon}$ or $\text{VP}_{\mathcal{F}}$? Run the simulation in both movement and stationary conditions, extract peak amplitudes at t^*, and regress against empirical N1/P3 amplitudes.

Q2. Is the discrete state space sufficient? The three-state model may collapse SURPRISED and RECOVER into a single transient state. If the recovery timecourse matters, consider replacing the discrete chain with a continuous belief $\pi_t \in [0,1]$ driven by γ.

Q3. What drives α_{eff}? In the model it is a fixed scalar. Biologically it may depend on the type of movement, training, or attentional state.

Q4. Can α_B and α_{eff} be fitted jointly to match both the kinematic perturbation magnitude ratio (movement/stationary) and the vertex potential amplitude ratio? This would provide a single quantitative test of the model.